



WT0132P4-A1 Datasheet



Version 2.4



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Revision History

Version	Date	Developed/ Changed Content	Modifier By	Auditor
V1.0	2024-9-12	First Creation	Pail	Louie
V1.1	2025-2-06	1.WT0132P4-A1 pin layout & description, modify pin 32 description. 2.Increase the cross-section of WT0132P4-A1 core board size.	Pail	Louie
V2.0	2025-3-14	Update Document Templates	Pail	Louie
V2.1	2025-4-09	1.Make additional explanations to some of the pins in 3.2Pin Description 2.Errata to 4.3 Recommended Working Conditions 3.Update dimension figure of WT0132P4-A1	Pail	Louie
V2.2	2025-5-29	1.Errata at Hardware Parameters	Pail	Louie
V2.3	2025-6-04	1.ESP32-P4 mains frequency 400 MHz corrected to 360 MHz, modified main chip architecture diagram.	Pail	Louie
V2.4	2025-7-28	1.Modify the pin layout diagram in 3.1 Pin Layout.	Pail	Louie



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1. Overview

1.1. Products Introduction

WT0132P4-A1 is an integrated NOR FLASH small core board based on Espressif ESP32-P4 chip designed by Wireless-Tag Technology Co., Limited. The core processor chip, ESP32-P4, can be stacked with 16MB or 32MB PSRAM in the package, and contains a high-performance (HP) system and a low-power (LP) system; the HP system adopts a RISC-V dual-core processor with a main frequency up to 360MHz, and contains a JPEG encoder/decoder, pixel-processing gas pedal, H.264 video encoder, and a MIPI interface; it has powerful image and voice processing capabilities. The HP system uses a RISC-V dual-core processor with up to 360MHz, including a JPEG coder/decoder, pixel processing accelerator, H.264 video encoder, and MIPI interface.

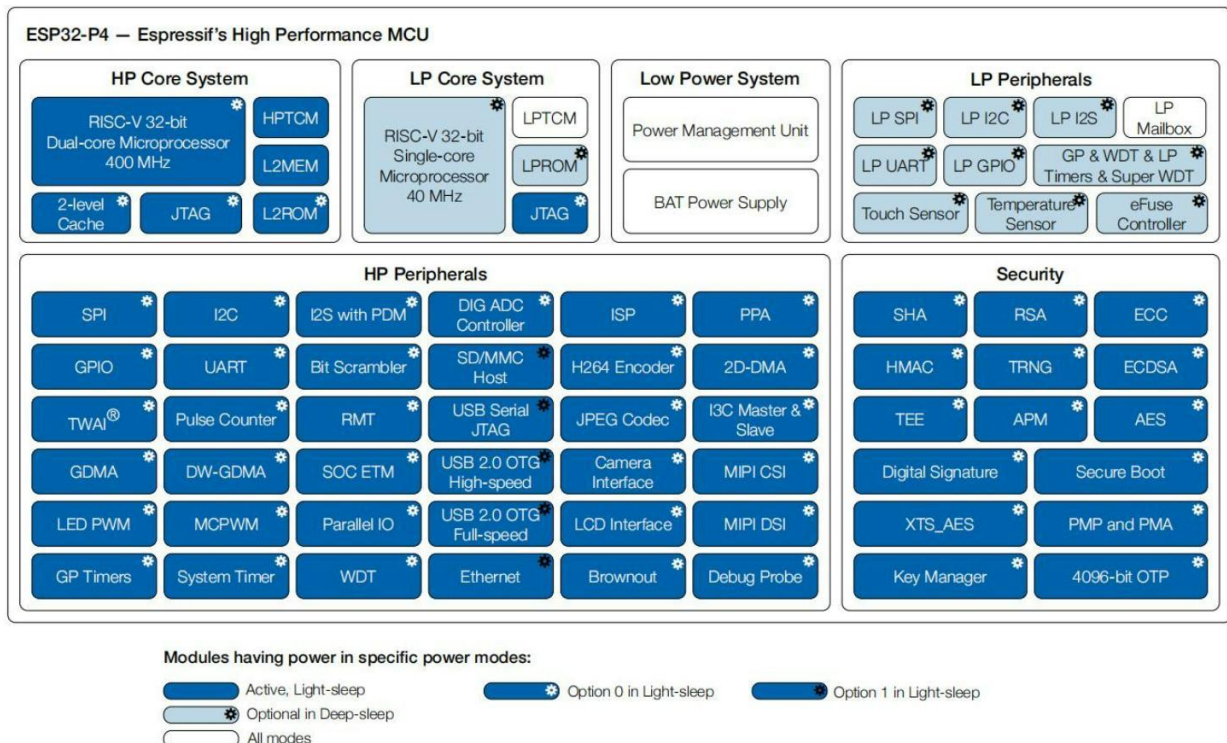


Figure 1: Main Chip Architecture Diagram

The WT0132P4-A1 series is available in two sizes, see the table below for more information.

WT0132P4-A1 Series Model Number Comparison

Part Number	Flash	Psram	Module Size (mm)
WT0132P4-A1-N16R16	16MB	16MB	25.00*20.00
WT0132P4-A1-N16R32	16MB	32MB	25.00*20.00

1.2.Product Features

- Dual-core 360 MHz high-frequency CPU
- 16 MB Flash and 16/32 MB Psram
- ESP32-P4 chip with full pinout
- Supports multiple multimedia interfaces
- Core board size is small, easy to hardware design
- Development materials are complete

1.3.Product Pictures



Figure 2:WT0132P4-A1(front)



Figure 3:WT0132P4-A1 (back)



Figure 4:WT0132P4-A1(front)



Figure 5:WT0132P4-A1(back)

1.4.Application Scenarios

- Smart Home
- Industrial Automation
- Consumer Electronics
- HMI Human Machine Interaction
- Electronic Robotics
- Camera Video Streaming
- USB Devices

2. Product Specification

2.1.Block Diagram

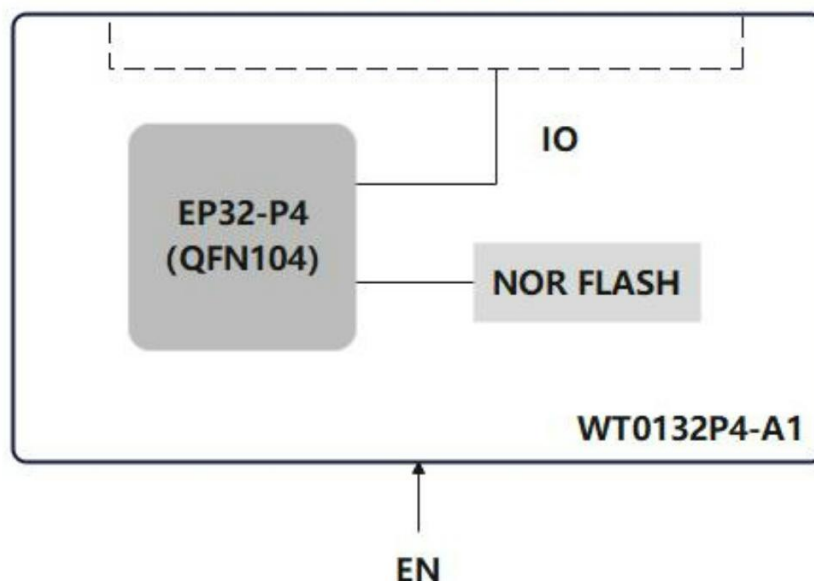


Figure 6: Block diagram of WT0132P4-A1

2.2.Hardware Parameters

CPU	CPU	ESP32-P4
	Core	32-bit RISC-V dual-core
	Main Frequency	360 MHz (HP Core) 40 MHz (LP Core)
Memory	ROM	128 KB HP ROM
		16 KB LP ROM



	SRAM	768 KB HP L2MEM
		32 KB LP SRAM
	Flash	16 MB
Peripheral Interface	GPIO	55
	SPI	2
	LP SPI	1
	UART	5
	LP UART	1
	I3C	1
	I2C	2
	LP I2C	1
	I2S	3
	LP I2S	1
	USB JTAG	1
	SDIO	1
	LED PWM	1
	MCPWM	2
	TWAI®Controller (compatible with ISO 11898-1)	3
	Hight-Speed USB 2.0 OTG	1
	Full-Speed USB 2.0 OTG	1
	100 Mbit Ethernet	1
	MIPI CSI-2	1
	MIPI DSI	1
	Parallel IO interface	1
	12-bit multi-channel ADC	2
	Temperature sensor	1
	Touch sensor	1
	Analog voltage comparator	1



	Brown-out detector	1
Image and Voice Processing Functionality	JPEG Codec	1
	PPA	1
	ISP	1
	H264 encoder	1

3. Pin Definitions

3.1.Pin Layout

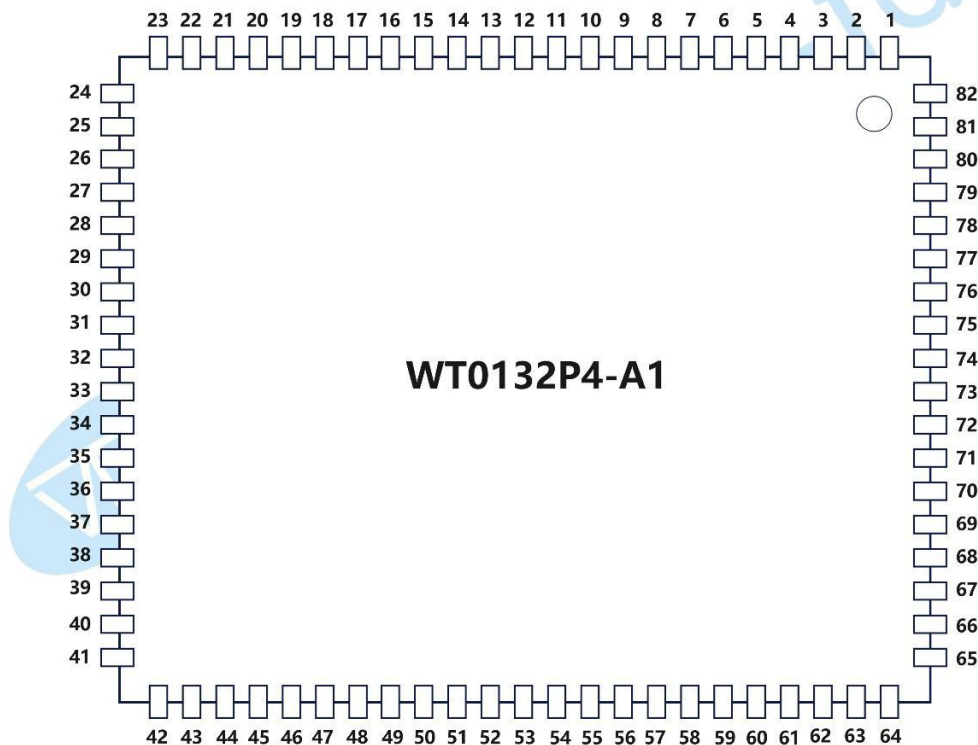


Figure 7:WT0132P4-A1 Pin Layout

3.2.Pin Description

Pin Definitions

No.	Name	Function
1	GND	GROUND
2	DSI_DATAP1	MIPI DSI PHY DATAP1
3	DSI_DATAN1	MIPI DSI PHY DATAN1
4	DSI_CLKN	MIPI DSI PHY CLKN
5	DSI_CLKP	MIPI DSI PHY CLKP



6	DSI_DATAP0	MIPI DSI PHY DATAP0
7	DSI_DATAN0	MIPI DSI PHY DATAN0
8	GND	GROUND
9	CSI_DATAN0	MIPI CSI PHY DATAN0
10	CSI_DATAP0	MIPI CSI PHY DATAP0
11	CSI_CLKP	MIPI CSI PHY CLKP
12	CSI_CLKN	MIPI CSI PHY CLKN
13	CSI_DATAN1	MIPI CSI PHY DATAN1
14	CSI_DATAP1	MIPI CSI PHY DATAP1
15	GND	GROUND
16	USB_DM	USB2 OTG PHY DM
17	USB_DP	USB2 OTG PHY DP
18	GND	GROUND
19	GPIO24	GPIO24, USB1P1_N0
20	GPIO25	GPIO25, USB1P1_P0
21	GND	GROUND
22	GPIO26	GPIO26, USB1P1_N1
23	GPIO27	GPIO27, USB1P1_P1
24	GPIO28	GPIO28, GPSPI SPI2 CS, EMAC PHY RXDV, DBG_PSRAM_D
25	GPIO29	GPIO29, GPSPI SPI2 D, EMAC PHY RXD0, DBG_PSRAM_Q
26	GPIO30	GPIO30, GPSPI SPI2 CK, EMAC PHY RXD1, DBG_PSRAM_WP
27	GPIO31	GPIO31, GPSPI SPI2 Q, EMAC PHY RXER, DBG_PSRAM_HOLD
28	GPIO32	GPIO32, I3CMST_SCL, GPSPI SPI2 HOLD. EMAC RMII CLK, DBG_PSRAM_DQ4
29	GPIO33	gpio33, i3cmst_sda, gpspi spi2 wp.



		EMAC PHY TXEN, DBG_PSRAM_DQ5
30	GPIO34	GPIO34, GPSPI SPI2 IO4, EMAC PHY TXD0, DBG_PSRAM_DQ6
31	GPIO35	GPIO35, GPSPI SPI2 IO5, EMAC PHY TXD1, DBG_PSRAM_DQ7 (IO35 pulls down into download mode)
32	ESP_LDO_VO4	Output POWER (Output voltage range 0.5~2.7V or 3.3V, maximum output current 0.2A)
33	GPIO36	GPIO36, GPSPI SPI2 IO6, EMAC PHY TXER, DBG_PSRAM_DQS0 (Default IO35, 36 pull-up to enter SPI Boot mode)
34	GPIO37	GPIO37, UART0_TXD, GPSPI SPI2 IO7 (Download port)
35	GPIO38	GPIO38, UART0_RXD, GPSPI SPI2 DQS (Download port)
36	GPIO39	GPIO39, SD1_CDATA0_PAD, REF_50M_CLK_PAD
37	GPIO40	GPIO40, SD1_CDATA1_PAD, GMAC_PHY_TXEN_PAD
38	GPIO41	GPIO41, SD1_CDATA2_PAD, GMAC_PHY_TXD0_PAD
39	GPIO42	GPIO42, SD1_CDATA3_PAD, GMAC_PHY_TXD1_PAD
40	GPIO43	GPIO43, SD1_CCLK_PAD, GMAC_PHY_TXER_PAD
41	VCC	POWER (5V input for core board power supply)
42	GND	GROUND
43	GPIO44	GPIO44, SD1_CCMD_PAD, GMAC_RMII_CLK_PAD
44	GPIO45	GPIO45, SD1_CDATA4_PAD, GMAC_PHY_RXDV_PAD
45	GPIO46	GPIO46, SD1_CDATA5_PAD, GMAC_PHY_RXD0_PAD
46	GPIO47	GPIO47, SD1_CDATA6_PAD, GMAC_PHY_RXD1_PAD
47	GPIO48	GPIO48, SD1_CDATA7_PAD, GMAC_PHY_RXER_PAD
48	GPIO49	GPIO49, GMAC_PHY_TXEN_PAD, ADC2_CHANNEL2
49	GPIO50	GPIO50, GMAC_RMII_CLK_PAD, ADC2_CHANNEL3



50	GPIO51	GPIO51, GMAC_PHY_RXDV_PAD, ADC2_CHANNEL4, ANA_COMP0
51	GPIO52	GPIO52, GMAC_PHY_RXD0_PAD, ADC2_CHANNEL5, ANA_COMP0
52	GPIO53	GPIO53, GMAC_PHY_RXD1_PAD, ADC2_CHANNEL6, ANA_COMP1
53	GND	GROUND
54	GPIO54	GPIO54, GMAC_PHY_RXER_PAD, ADC2_CHANNEL7, ANA_COMP1
55	GPIO2	GPIO2, MTCK, LP_GPIO 2, TOUCH_CHANNEL0
56	GPIO3	GPIO3, MTDI, LP_GPIO 3, TOUCH_CHANNEL1
57	GPIO4	GPIO4, MTMS, LP_GPIO4, TOUCH_CHANNEL2
58	GPIO5	GPIO5, MTDO, LP_GPIO5, TOUCH_CHANNEL3
59	GPIO6	GPIO6, SPI2_HOLD_PAD, LP_GPIO6, TOUCH_CHANNEL4
60	GPIO7	GPIO7, SPI2_CS_PAD, LP_GPIO7, TOUCH_CHANNEL5
61	GPIO8	GPIO8, UART0_RTS_PAD, SPI2_D_PAD , LP_GPIO8, TOUCH_CHANNEL6
62	GPIO9	GPIO9, UART0_CTS_PAD, SPI2_CK_PAD, LP_GPIO9, TOUCH_CHANNEL7
63	GPIO10	GPIO10, UART1_TXD_PAD, SPI2_Q_PAD, LP_GPIO10, TOUCH_CHANNEL8
64	GND	GROUND
65	GPIO11	GPIO11, UART1_RXD_PAD, SPI2_WP_PAD, LP_GPIO11, TOUCH_CHANNEL9
66	GPIO12	GPIO12, UART1_RTS_PAD , LP_GPIO12, TOUCH_CHANNEL10
67	GPIO13	GPIO13, UART1_CTS_PAD, LP_GPIO13, TOUCH_CHANNEL11
68	GPIO14	GPIO14, LP_GPIO14, LP_UART_TXD_PAD,

		TOUCH_CHANNEL12
69	GPIO15	GPIO15, LP_GPIO15, LP_UART_RXD_PAD, TOUCH_CHANNEL13
70	CHIP_PU	Enable P4 chip (internal 10K pull-up)
71	GPIO0	GPIO0, LP_GPIO0, XTAL_32K_N
72	GPIO 1	GPIO1, LP_GPIO1, XTAL_32K_P
73	GND	GROUND
74	GPIO16	GPIO16, ADC1_CHANNEL0
75	GPIO17	GPIO17, ADC1_CHANNEL1
76	GPIO18	GPIO18, ADC1_CHANNEL2
77	GPIO19	GPIO19, ADC1_CHANNEL3
78	GPIO20	GPIO20, ADC1_CHANNEL4
79	GPIO21	GPIO21, ADC1_CHANNEL5
80	GPIO22	GPIO22, ADC1_CHANNEL6
81	GPIO23	GPIO23, ADC1_CHANNEL7, REF_50M_CLK_PAD
82	GND	GROUND

3.3.Startup Item Configuration

3.3.1. Strapping Pins

The chip allows for configuring the following boot parameters through strapping pins and eFuse bits at power-up or a hardware reset, without microcontroller interaction.

- **Chip Boot Mode**

- Strapping pin: GPIO35, GPIO36, GPIO37 and GPIO38

- **ROM Message Printing**

- Strapping pin: GPIO36

- eFuse bit: EFUSE_UART_PRINT_CONTROL

- **JTAG Signal Source**

- Strapping pin: GPIO34

– eFuse bit: EFUSE_DIS_PAD_JTAG, EFUSE_DIS_USB_JTAG and EFUSE_JTAG_SEL_ENABLE

The default values of all the above eFuse bits are 0, which means that they are not burnt. Given that eFuse is one-time programmable, once an eFuse bit is programmed to 1, it can never be reverted to 0.

The default values of the strapping pins, namely the logic levels, are determined by pins' internal weak pull-up/pull-down resistors at reset if the pins are not connected to any circuit, or connected to an external high-impedance circuit.

Default Configuration of Strapping Pin

Strapping Pin	Default Configuration	Value
GPIO34	Floating	-
GPIO35	Weak pull-up	1
GPIO36	Floating	-
GPIO37	Floating	-
GPIO38	Floating	-

To change the bit values, the strapping pins should be connected to external pull-down/pull-up resistors. If the ESP32-P4 is used as a device by a host MCU, the strapping pin voltage levels can also be controlled by the host MCU.

All strapping pins have latches. At system reset, the latches sample the bit values of their respective strapping pins and store them until the chip is powered down or shut down. The states of latches cannot be changed in any other way. It makes the strapping pin values available during the entire chip operation, and the pins are freed up to be used as regular IO pins after reset.

3.3.2. Chip Boot Mode Control

GPIO35 ~ GPIO38 control the boot mode after the reset is released.

Boot Mode	GPIO35	GPIO36	GPIO37	GPIO38
SPI Boot*	1*	Any value	Any value	Any value



Joint Download Boot	0	1	Any value	Any value
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*marks the default value and configuration.

Joint Download Boot mode supports the following download methods:

- USB Download Boot:
 - USB-Serial-JTAG Download Boot
 - USB 2.0 OTG Download Boot
- UART Download Boot
- SPI Slave Download Boot

3.3.3. ROM Messages Printing Control

During the boot process, the messages by the ROM code can be printed to:

- **(Default) UART0 and USB Serial/JTAG controller**
- USB Serial/JTAG controller
- UART0

EFUSE_UART_PRINT_CONTROL and GPIO36 control ROM messages printing to UART0 as shown in Table.

UART0 Code Printing	EFUSE_UART_PRINT_CONTROL	GPIO36
Enabled*	0*	Ignored
	1	0
	2	1
Disabled	1	1
	2	0
	3	Ignored

*marks the default value and configuration.

EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT controls the printing to USB Serial/JTAG controller as shown in Table.

USB Serial/JTAG ROM Code Printing	EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT
Enabled*	0*
Disabled	1

*marks the default value and configuration.

4. Electrical Characteristics

4.1. Absolute Maximum Limit Value

Exceeding the absolute maximum ratings may result in permanent damage to the device. This is an emphasized rating only and does not address the functional operation of the device under these or other conditions beyond those indicated in these specifications. Prolonged exposure to absolute maximum rating conditions may affect module reliability.

4.2. Power Consumption Characteristics

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4.3. Recommended Working Conditions

Parameter	Description	Min	Typ	Max	Unit
VCC	Power pin voltage	4.8	5	5.5	V
IVCC	Supply current from external power supply	-	1	-	A
TA	Operating Temperature	-40	-	85	°C

5. WT0132P4-A1 Schematic

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Figure 8: WT0132P4-A1 Schematic

6. WT0132P4-A1 Dimensions

The following figure shows the top view and front view of WT0132P4-A1 with a tolerance of ± 0.2 mm.

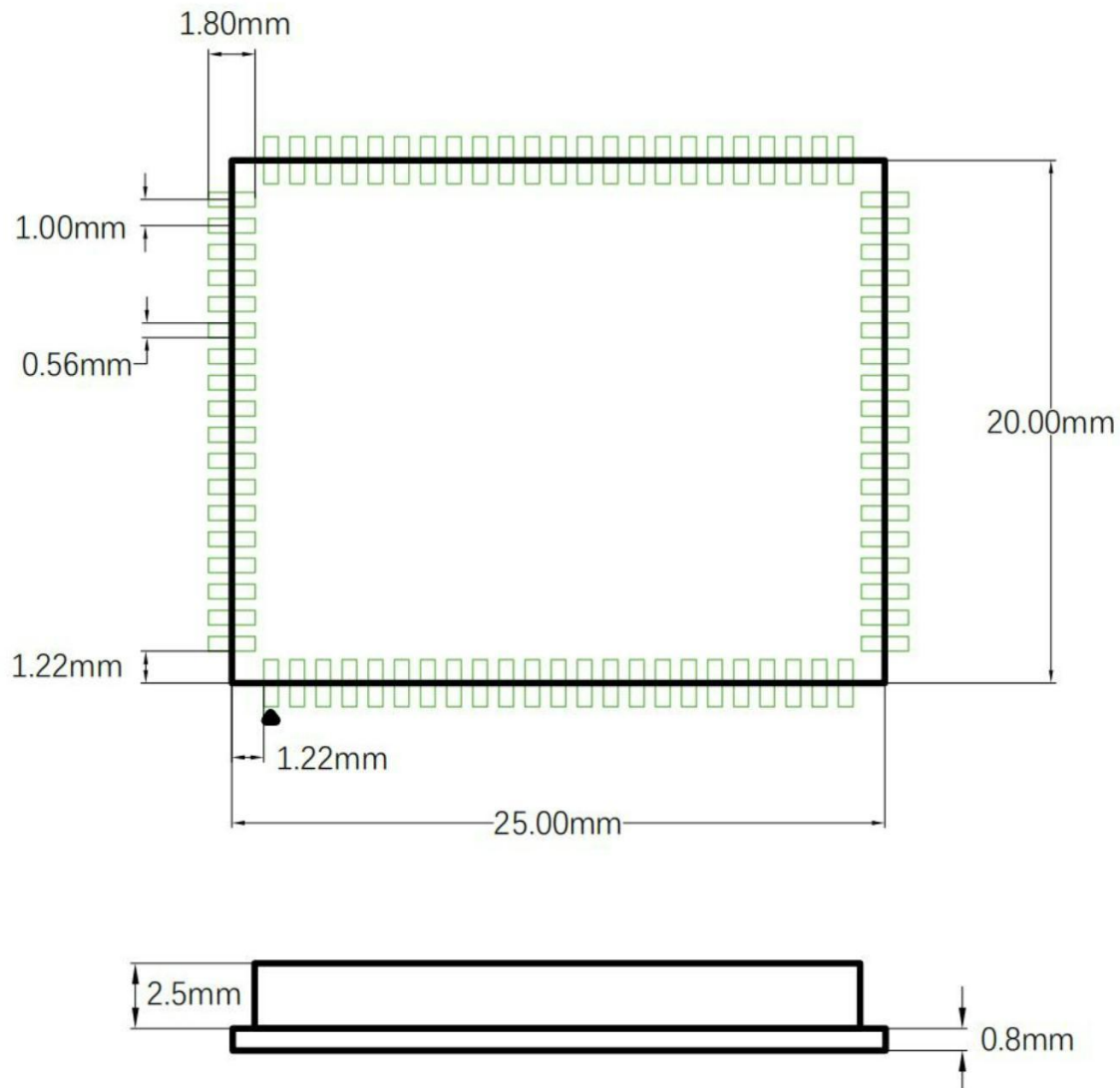


Figure 9: Dimension figure of WT0132P4-A1

7. Storage Condition

Prerequisite	Parameters
Storage condition	Non-condensing atmosphere < 40°C /90 %RH in sealed MBBs
Conditions of use	168 hours at 25 ± 5°C , 60 % RH.
Moisture sensitivity	3 levels

8. Reflow Soldering Curve

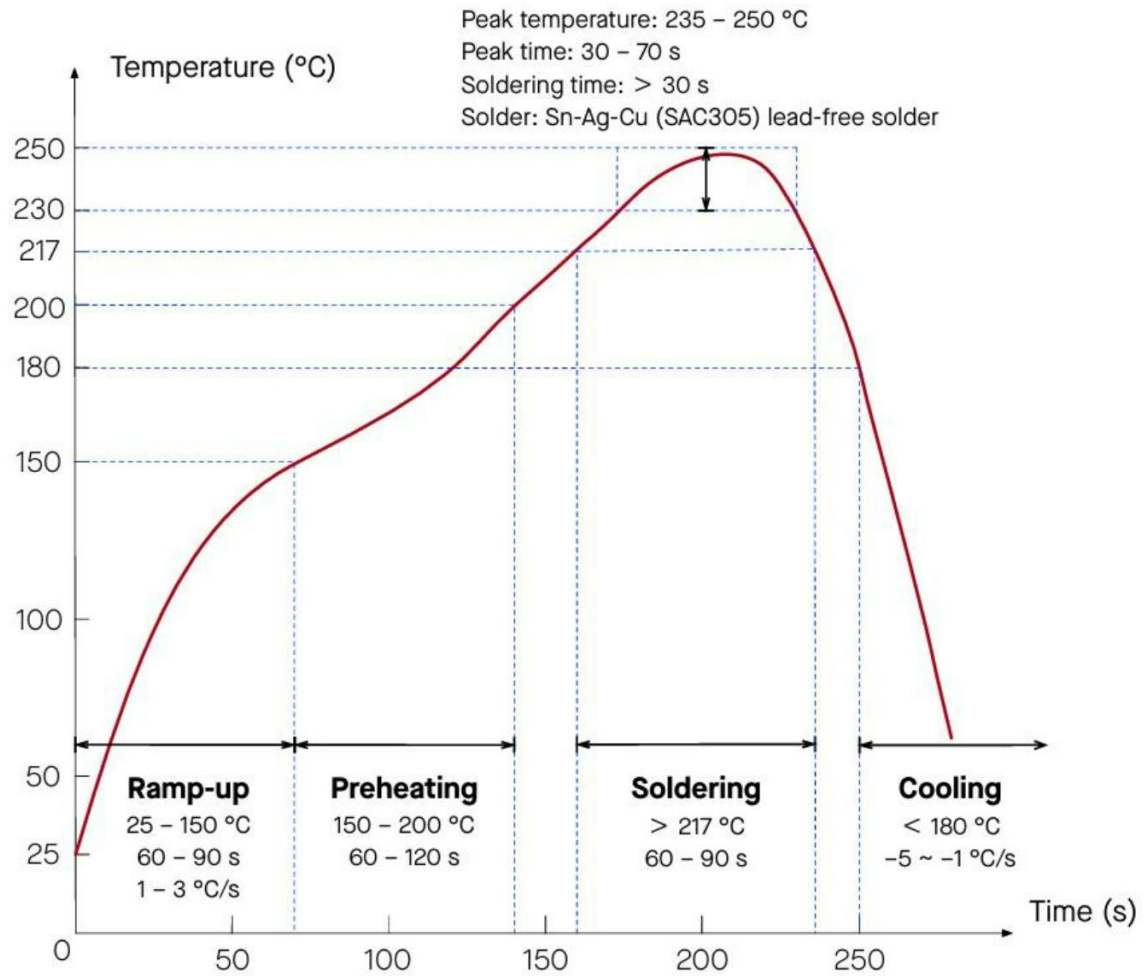


Figure 10: Reflow Soldering Temperature Curve